

Asaf Iron-Jobes EE446 HW2

Part 3 report output values

```
... WARNING:absl:At this time, the v2.11+ optimizer `tf.keras.optimizers.Adam` runs slowly on M1/M2 Macs, please use the legacy Keras optimizer
Run and Debug (⌘⇧D) jrtial"
```

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	2496
dense_1 (Dense)	(None, 32)	2080
dense_2 (Dense)	(None, 16)	528
dense_3 (Dense)	(None, 2)	34

=====
Total params: 5138 (20.07 KB)
Trainable params: 5138 (20.07 KB)
Non-trainable params: 0 (0.00 Byte)

Epoch	1/10
355/355	[=====] - 1s 844us/step - loss: 0.0851 - accuracy: 0.9730 - val_loss: 0.0205 - val_accuracy: 0.9942
Epoch 2/10	
355/355	[=====] - 0s 633us/step - loss: 0.0153 - accuracy: 0.9950 - val_loss: 0.0132 - val_accuracy: 0.9951
Epoch 3/10	
355/355	[=====] - 0s 620us/step - loss: 0.0105 - accuracy: 0.9963 - val_loss: 0.0092 - val_accuracy: 0.9960
Epoch 4/10	
355/355	[=====] - 0s 616us/step - loss: 0.0083 - accuracy: 0.9969 - val_loss: 0.0128 - val_accuracy: 0.9970
...	
Epoch 9/10	
355/355	[=====] - 0s 625us/step - loss: 0.0047 - accuracy: 0.9984 - val_loss: 0.0113 - val_accuracy: 0.9983
Epoch 10/10	
355/355	[=====] - 0s 615us/step - loss: 0.0045 - accuracy: 0.9986 - val_loss: 0.0056 - val_accuracy: 0.9981

Output is truncated. View on a scrollable element or open in a text editor. Adjust cell output settings.

```
788/788 [=====] - 0s 292us/step
```

	precision	recall	f1-score	support
attack	1.00	1.00	1.00	11787
normal	1.00	1.00	1.00	13408
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

```
[[11771 16]
 [ 34 13374]]
```

Part 4 report outputs

Original model size: 100.01 KB
Quantized model size: 9.54 KB

```
] ✓ 0.1s

INFO: Created TensorFlow Lite XNNPACK delegate for CPU.

      precision    recall  f1-score   support

   attack         1.00      1.00      1.00     11787
   normal         1.00      1.00      1.00     13408

 accuracy              1.00     25195
 macro avg           1.00      1.00      1.00     25195
weighted avg           1.00      1.00      1.00     25195

[[11770    17]
 [   34 13374]]
```

Part 5:

Actual Class: 118
Sample #: 7
Predicted Class: 0
Actual Class: 118
Sample #: 8
Predicted Class: 0
Actual Class: 97
Sample #: 9
Predicted Class: 1
Actual Class: 108
Sample #: 0
Predicted Class: 1
Actual Class: 1
Sample #: 1
Predicted Class: 1
Actual Class: 1
Sample #: 2
Predicted Class: 0
Actual Class: 0
Sample #: 3
Predicted Class: 0
Actual Class: 0
Sample #: 4
Predicted Class: 1
Actual Class: 1
Sample #: 5
Predicted Class: 1
Actual Class: 73
Sample #: 6
Predicted Class: 0
Actual Class: 110
Sample #: 7
Predicted Class: 0
Actual Class: 118
Sample #: 8
Predicted Class: 0
Actual Class: 97
Sample #: 9
Predicted Class: 1
Actual Class: 108

Setup started
Model loaded
Arena used bytes: 1084
Sample #: 0
Predicted Class: 1
Actual Class: 1
Sample #: 1
Predicted Class: 1
Actual Class: 1
Sample #: 2
Predicted Class: 0
Actual Class: 0
Sample #: 3
Predicted Class: 0
Actual Class: 0
Sample #: 4
Predicted Class: 1
Actual Class: 1